

Question

Using MATLAB/Scilab, design a quarter-wave transformer for matching a $90\ \Omega$ load to a $50\ \Omega$ line at 2.4 GHz. Calculate the transformer impedance and verify the input impedance after matching.

Scilab Code:

```
1  clc;
2  clear;
3  // Given values
4  Z0 = 50; // Characteristic impedance of line
5  ZL = 90; // Load impedance
6  f = 2.4e9; // Frequency in Hz
7  c = 3e8; // Speed of light
8
9  // Calculate transformer impedance
10 Zt = sqrt(Z0 * ZL);
11
12 // Calculate wavelength
13 lambda = c / f;
14
15 // Calculate quarter-wave length
16 l = lambda / 4;
17
18 // Calculate input impedance
19 Zin = (Zt^2) / ZL;
20
21 // Display results
22 disp("QUARTER-WAVE TRANSFORMER");
23 disp("-----");
24
25 mprintf("Line impedance Z0 = %.2f Ohms\n", Z0);
26 mprintf("Load impedance ZL = %.2f Ohms\n", ZL);
27 mprintf("Frequency = %.2f GHz\n", f/1e9);
28 mprintf("Transformer impedance Zt = %.2f Ohms\n", Zt);
29 mprintf("Wavelength = %.4f m\n", lambda);
30 mprintf("Quarter-wave length = %.4f m\n", l);
31 mprintf("Input impedance Zin = %.2f Ohms\n", Zin);
32 if abs(Zin - Z0) < 0.01 then
33     disp("Matching verified successfully.");
34 else
35     disp("Matching not achieved.");
36 end
37
```

Line 31, Column 50.

Expected Output

QUARTER-WAVE TRANSFORMER

Line impedance Z0 = 50.00 Ohms

Load impedance $Z_L = 90.00 \text{ Ohms}$

Frequency = 2.40 GHz

Transformer impedance $Z_t = 67.08 \text{ Ohms}$

Wavelength = 0.1250 m

Quarter-wave length = 0.0313 m

Input impedance $Z_{in} = 50.00 \text{ Ohms}$

Matching verified successfully.

OUTPUT:

```
Scilab 2025.1.0 Console
"QUARTER-WAVE TRANSFORMER"
"-----"
Line impedance  $Z_0 = 50.00 \text{ Ohms}$ 
Load impedance  $Z_L = 90.00 \text{ Ohms}$ 
Frequency = 2.40 GHz
Transformer impedance  $Z_t = 67.08 \text{ Ohms}$ 
Wavelength = 0.1250 m
Quarter-wave length = 0.0312 m
Input impedance  $Z_{in} = 50.00 \text{ Ohms}$ 
"Matching verified successfully."
-->
```

Result

The quarter-wave transformer impedance required to match the **90 Ω loads** to the **50 Ω transmission line** is **67.08 Ω** . At **2.4 GHz**, the quarter-wave transformer length is **3.125 cm**. The calculated input impedance is **50 Ω** , confirming that the load is successfully matched to the transmission line.

